

D1.2 We are separating a mixture of methanol and water in a flash drum at 1 atm pressure. Equilibrium data are listed in Table 2.8.

a. Feed is 60.0 kmol/h methanol, and 40.0% of the feed is vaporized. What are the vapor and liquid mole fractions and flow rates? Feed rate is 100.0 kmol/h.

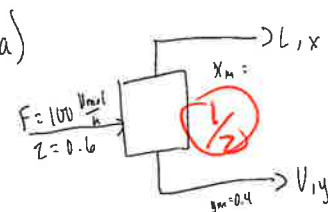
b. Repeat part a for a feed rate of 1000.0 kmol/h.

c. If the feed is 50.0 mol% methanol and we desire a liquid product that is 20.0 mol% methanol, what V/F must be used? For a feed rate of 1000.0 kmol/h, find product flow rates and compositions.

d. We are operating the flash drum so that the liquid mole fraction is 45 mol% methanol, $L = 1500.0$ kmol/h, and $V/F = 0.20$. What must the flow rate and composition of the feed be?

*graph
sketch
on back

2.D1a)



$$\frac{41.9 - 76.5}{10 - 9} = \frac{40 - 36.5}{x - 9}$$

$$7 = 5.3(x - 9)$$

$$7 = 5.3x - 47.4$$

$$49.4 = 5.3x$$

$$x = 9.32$$

$$F = L + V$$

$$\frac{V}{F} = 0.4 \rightarrow V = 0.4F = 40 \frac{\text{kmol}}{\text{h}}$$

$$100 = L + 40 \rightarrow L = 60 \frac{\text{kmol}}{\text{h}}$$

$$FZ = Lx + Vy$$

$$7 = 0.6$$

$$y = y_1 + \frac{(x - x_1)(y_2 - y_1)}{(x_2 - x_1)}$$

$$(100)(0.6) = 60x + 40y \rightarrow 60 = 60x + 40y$$

$$60 - 60x = 40y$$

$$y = 1.5 - 1.5x$$

$$y = 72.9 + \frac{(x - 90)(77.9 - 72.9)}{(52 - 90)}$$

$$y = 72.9 + 1.5(x - 40) = 72.9 + 0.5x - 20$$

$$y = 0.5x + 52.9$$

$$y = 0.5x + 52.9 \rightarrow \text{optimal line}$$

$$0.5x + 52.9 = 150 - 1.5x$$

$$2x = 97.1$$

$$x = 48.55\% \text{ liquid mole fraction } (x = 0.4855)$$

$$y = 150 - 1.5(48.55) = 72.18\% \text{ vapor mole fraction } (y = 0.7218)$$

$$FZ = Lx + Vy \rightarrow (100)(0.6) = (60)(0.4855) + (40)(0.7218)$$

$$60 = 60.002$$

$$x, y = (0.4855, 0.7218)$$

2D1b)

$$FZ = Lx + Vy \rightarrow FZ - Lx = Vy \rightarrow y = -\frac{L}{V}x + \frac{FZ}{V}$$

$$\frac{V}{F} = 0.4 \rightarrow V = 0.4F = 40 \frac{\text{kmol}}{\text{h}} \rightarrow F = L + V \rightarrow L = 1500 - 600 = 900 \frac{\text{kmol}}{\text{h}}$$

$$y = -\frac{900}{600}x + \frac{1500}{600}Z = -1.5x + 2.5Z = -1.5x + 1.5 \text{ (same result from part a)}$$

$$x = 0.4855 \text{ liquid fraction}$$

$$y = 0.7218 \text{ vapor fraction}$$

$$x, y = (0.4855, 0.7218)$$

$$\text{The mole fractions would stay the same since only the flow rate changed and not the slope } (-\frac{L}{V}), \frac{FZ}{V}, \text{ or } Z. \text{ We know from this feed rate doesn't affect } \text{Vapor in equilibrium.}$$

2D1c)

$$F = V + L \quad FZ = Vy + Lx \quad 1 = \frac{V}{F} + \frac{L}{F} \quad \frac{L}{F} = (1 - \frac{V}{F})$$

$$Z = \frac{V}{F}y + \frac{L}{F}x \rightarrow Z = \frac{V}{F}y + (1 - \frac{V}{F})x = \frac{V}{F}y + x - \frac{V}{F}x$$

$$Z = x + \frac{V}{F}(y - x) \rightarrow \frac{Z - x}{y - x} = \frac{V}{F}$$

$$\frac{V}{F} = \frac{0.3 - 0.2}{0.579 - 0.2} = 0.2638$$

$$\frac{V}{F} = 0.2638 \rightarrow V = 0.2638(1000) = 263.8 \frac{\text{kmol}}{\text{h}}$$

$$L = F - V = 1000 - 263.8 = 736.2 \frac{\text{kmol}}{\text{h}}$$

$$x = 0.2 \text{ liquid mole fraction}$$

$$y = 0.579 \text{ vapor mole fraction (given \& table)}$$

$$x = 0.45 \quad L = 1500 \frac{\text{kmol}}{\text{h}} \quad \frac{V}{F} = 0.2 \rightarrow V = 0.2F$$

$$F = L + V \quad FZ = Lx + Vy$$

$$L = F - V = F - 0.2F = F(1 - 0.2)$$

$$F = \frac{L}{0.8} = \frac{1500}{0.8} = 1875 \frac{\text{kmol}}{\text{h}}$$

$$Z = \frac{Lx + Vy}{F} = \frac{(1500)(0.45) + (375)(0.734)}{1875} = 0.5108$$

$$\text{mole fraction methanol}$$

$$y = 72.9 + \frac{(45 - 90)(77.9 - 72.9)}{50 - 40} = 75.4\%$$

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$$\text{mole fraction methanol}$$

2b)

$$y = -x + 0.8 \quad Z = 0.4$$

$$Qx = 0.4 \quad (x, y) = (0.4, 0.4)$$

$$y = 0.72 \quad Z = 0.4$$

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2c)

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